

Some Interesting Facts

Consider the following:

- Intel saw a 1 percent drop in profits
- PC sales have stagnated. Mobile sales overshot PC sales.
- It is estimated that there will be over 600 million smartphones in 2014
- There are currently around 670 million Windows users and only 40 million Windows 8 users
- The gaming industry saw its first decline in nearly 10 years.
- Intel is trying to get into mobile devices with its Atom processors (and failing)
- ARM is trying to compete with x86 and has announced its first 64-bit chips
- AMD plans to produce ARM-based Opeterons over the next few years.
- Rumours abound that Apple will shift to ARM processors for all its devices.
- Microsoft is desperately trying to get its ARM ecosystem up and running.

Deja Vu

The tale of the IBM-PC is a cautionary one, a tale that Wintel needs to revisit if it has to survive, or should have. When the first IBM-PC was launched around 1980, there were a large variety of PCs around and no set platform for developers to work on. Over time, the IBM-PC and its successors, running MS-DOS, became the industry standard as far as PCs went. These were the first, true precursors to the current Wintel devices, featuring a Microsoft OS running on x86 architecture. The success of the design led to the development of IBM-PC compatible clones that were sold by every other manufacturer including HP. These were the heydays of IBM; every PC in the world wanted to be IBM compatible and they were in a position to define and dominate home computing. Not unlike the Wintel that we see today.

Ultimately, IBM ended up sounding its own death knell when it released the IBM-P/S2. The PC was different from all earlier IBM designs and crucially, was not IBM PC compatible. The programs available at the time were mostly incompatible and developers would have again had to compile programs for this new architecture all over again. The sales of this PC declined, clones flourished and IBM's PC died. Microsoft, with its push for the ARM architecture and Windows RT, is doing exactly this. With the release of the RT Microsoft rendered unusable the (possibly) billions of apps that are available for the x86 platform and is forcing the developers to start from scratch all over again. Not a good sign. Why would anyone want to go for a completely new ecosystem with a handful of apps when there is an already existing ecosystem (Apple and Android), that come with hundreds of thousands of them?

ecosystem to speak of and the Atom is too power hungry for the performance on offer. On the other hand, ARM is too slow to support a full-blown desktop OS and while way more efficient, lacks the freedom that x86 offers.

A RISCy future

The only future is some sort of middle ground between these differing architectures and the company that reaches there first, wins. Intel has been too dependent on Microsoft and even though Apple has also switched to Intel

CPUs, there are very persistent rumours floating around that Apple might switch to ARM for all future devices. The PC has been relegated to the state of a true workstation now, for someone who needs that amount of horsepower, such as a gamer, graphic designer, etc. Even AMD is planning a switch to ARM-based Opeterons in the future which, if designed right, might sound the final death knell for x86.

If the future is RISC then both intel and Microsoft need to move really fast to come up with a compatible architecture

and OS but the fact remains, that unless something is done soon, both will have lost the initiative and might end up sharing the same fate as Nokia. In this cutthroat market, too-little-too-late is just not going to cut it anymore. People want

complete ecosystems and functional devices that will not disrupt their lives and still provide them with a better experience than they are used to.

Is Wintel dead?

No. The current PC market has received a severe shaking up, the dust is yet to settle and no true winners (there have been losers) have been spotted yet. The current ARM-based Windows RT is disappointing and unfinished; intel's Atom is RISC architecture masquerading as CISC and is woefully underpowered when it comes to graphics. Wintel is not the monopoly that it once was and needs to adapt fast to keep its position.

The current state of Wintel's income and market share has jolted them from their complacency and made them face the fact that the PC has changed, but they need to understand that compromises and arm twisting (pardon the pun) are not acceptable anymore. If Wintel can do that, redefine computing on their terms, provide a compelling experience for users, give them a platform that they can trust and use, they will find themselves back at the top instead of at the bottom desperately trying to play catch up. **d**



The last of its kind?